The exceptional Coxeter censuses

Human Compression project: F006 recovery

Working manuscript; distinct review pending

We allow an adjacent product of order two in a string C-marking. Consequently its string diagram need not be connected. Counts below identify a marking with its reverse; parenthesized numbers count self-dual classes. The complete proofs of enumeration by stabilizers of initial strings and of its implementation through double cosets are acquired prerequisites. The exact finite executions specified below are additional premises. The classical finite Coxeter presentations and orders, and exact GAP group operations, are retained inputs throughout.

Theorem 1. *The complete census for the following five finite Coxeter groups is given in the table. Every omitted rank has zero classes.*

	3	4	5	6	7	*total*
F_4	3(0)	2(1)	0	0	0	5(1)
H_4	45(2)	14(4)	0	0	0	59(6)
E_6	87(12)	50(4)	10(2)	0	0	147(18)
E_7	1577(10)	1987(0)	741(0)	167(0)	18(0)	4490(10)
E_8	6746(117)	3584(22)	986(2)	310(0)	63(1)	11689(142)

All markings for F_4, H_4, E_6, E_8 are connected. The connected part of the E_7 census is

	3	4	5	6	7	*total*
E_7	1577(10)	1525(0)	465(0)	95(0)	0	3662(10).

Thus the respective maximum ranks are $4, 4, 5, 7, 7$ under the stated convention; the connected maximum for E_7 is 6.

Proof. The models are exact reflection permutations on finite root sets. The simple reflections satisfy the appropriate Coxeter relations, and their generated groups have orders

$$1152, \quad 14400, \quad 51840, \quad 2903040, \quad 696729600.$$

The classical presentations therefore give surjections from the specified Coxeter groups onto the models. Equality with the classical orders makes these maps isomorphisms. The complete constructors, root coordinates, permutations and identity checks are retained finite evidence.

For F_4, H_4, E_6, the enumeration uses the full automorphism action on the complete involution set. For E_7, the exact guard

$$|E_7 : Z(E_7)| = |\operatorname{Aut}(E_7)|$$

allows conjugation by the group itself. Its centralizers are the inverse images of the pointwise automorphism stabilizers. The central kernel acts trivially and changes none of their orbits.

For E_8, use the acquired construction on two permutation domains. In the original degree-240 model, let z be the nonidentity central element and let χ be permutation parity. The finite guards establish $|z| = 2$ and $\chi(z) = 0$. Thus $\phi(g) = z^{\chi(g)}g$ is an involutory automorphism. Represent g on two copies of the original domain by g and $\phi(g)$, respectively. The permutation

interchanging the copies implements ϕ. Adjoining it gives a group A normalizing the degree-480 copy of E_8. The exact guard gives

$$|A| = 1393459200, \quad |C_A(E_8)| = 2, \quad |A : C_A(E_8)| = |\operatorname{Aut}(E_8)| = 696729600.$$

Hence conjugation by A supplies the full automorphism group. The acquired proof justifies its use at every prefix stabilizer and in every reversal test; a full action is checked here, rather than presumed from the form of the central automorphism.

The E_8 execution enumerates candidate orbits through classes in the current centralizer and the corresponding double cosets. This is the acquired exact implementation of the same stabilizer search. It also discards central involutions in the derived subgroup. To justify this last step, if a distinguished central involution t_i occurred in a C-marking, the intersection property would give

$$G = \langle t_i \rangle \times \langle t_j : j \neq i \rangle.$$

Every commutator lies in the second factor, whereas t_i does not. Thus such a mark lies outside G'. The subgroup $Z(G) \cap G'$ is characteristic, so deleting its involutions preserves the candidate-domain invariance and excludes no valid marking. The acquired completion test uses the subgroup generated by all remaining eligible elements, as calculated by the normal closure of their complete class representatives.

The complete ordered counts at successive nonempty ranks are $6, 3$ for F_4; $88, 24$ for H_4; $162, 96, 18$ for E_6; $3144, 3974, 1482, 334, 36$ for E_7; and $13375, 7146, 1970, 620, 125$ for E_8. The corresponding self-dual counts are $(0, 1)$, $(2, 4)$, $(12, 4, 2)$, $(10, 0, 0, 0, 0)$ and $(117, 22, 2, 0, 1)$. The exact test for an automorphism sending a tuple to its reverse gives these fixed-point counts. Reversal is an involution on the ordered automorphism classes, so N ordered classes and S fixed classes give $(N + S)/2$ classes modulo duality. This proves the first table.

Restricting to tuples whose adjacent products have order greater than two leaves every tuple except the disconnected E_7 markings. In that group it leaves $3144, 3050, 930, 190, 0$ ordered tuples at ranks three through seven, with self-dual counts $10, 0, 0, 0, 0$. The same reversal formula proves the second table. The five complete executions cover every rank through $\lfloor \log_2 |G| \rfloor$, namely $10, 13, 15, 21, 29$. They retain the actual generating tuples, action and model generators, all rank summaries, and the final exhaustion records. The acquired enumeration proof establishes coverage and uniqueness, including every zero row. The asserted maxima follow from the last nonzero ranks. □

A separate rank-seven witness. The first retained rank-seven E_7 tuple has adjacent product orders $(6, 6, 4, 3, 3, 2)$. Its final mark is central of order two. The other six marks generate a subgroup of order 1451520 intersecting that central subgroup trivially; together they generate the reflection group of order 2903040. A separate execution verifies the intersection property directly for all 16384 pairs of subsets of the seven marks. The literal tuple and execution are retained. This provides a witness independently of the census counts. Its diagram is disconnected because the last adjacent product has order two.

Comparison with the source. Chen and Rowley, *Abstract Regular Polytopes of Finite Irreducible Coxeter Groups*, Electronic Journal of Combinatorics 32(4) (2025), P4.18, Table 1, physical page 12, gives exactly the connected counts in Theorem 1. The displayed definition on page 1 permits disconnected strings, and page 2 expressly discusses them for B_n. Under that literal convention, the E_7 clause of Theorem 3(ii) requires the rank-seven correction. Its rank-six assertion is correct with connectedness imposed. The independently acquired $I_2(m)$ and H_3 component records the analogous scope corrections for those groups.

The original authors supply the target results and published connected counts. The enumeration method follows the established Hartley–Hulpke and Leemans–Mixer approach, with

attribution and the precise executable tests given in the acquired proofs. The two-domain implementation and central exclusion use standard group constructions; no historical priority is claimed. The new executions recover the printed counts and retain disconnected cases under the source's literal definition. They use GAP as an explicit software premise and are not Lean certificates. The present manuscript makes no open-conjecture or numerical compression claim. The acquired uniform D_n proofs and the small-group component are required for a complete F006 recovery.